

ECON 282E — Homework 1, Part B

Continuous Choice, Stochastic Shocks, and Accuracy

Foundations of Macroeconomics
University of California, Riverside

Out: Thursday, October 15, 2026 • Due: Thursday, October 22, 2026

What to hand in. One PDF, your code, and `AI_LOG.md` — together with Part A, in a single submission. Question 3(d) feeds the anonymous accuracy leaderboard.

Continuity with Part A. Questions 2(d) and 2(e) extend the same deterministic model and the same solver. Question 3 changes the calibration to a standard quarterly one and adds a shock.

Question 2, continued. Getting off the grid (40 points)

Same model and calibration as Part A: $u = \ln c$, $c + k' = Zk^\alpha$, $\delta = 1$, $\alpha = 0.36$, $\beta = 0.95$, $Z = 1$.

- (d) **(20 pts) Continuous choice.** Replace the grid search with a maximization over the *interval*, using golden section (or Brent) and evaluating the interpolated value function off the grid. Keep $N = 200$.
- (i) Report the maximum relative policy error and the maximum unit-free Euler residual, evaluated at 500 capital levels that are *not* grid points, against your Part A 2(a) results.
 - (ii) Report the number of objective evaluations and the wall time. How do they compare?
 - (iii) Which interpolation scheme did you use for $\hat{v}$, and why does the choice matter for whether golden section is licensed at all?
- (e) **(20 pts) A second road.** Solve the same model again by *either* time iteration on the Euler equation *or* the endogenous grid method. Then produce one table comparing all four solvers — Part A 2(a), Part A 2(c), and 2(d) and 2(e) here — on iterations, wall time, maximum relative policy error and maximum off-grid Euler residual. Which solver would you use in practice, and for what reason?

Question 3. A stochastic, quarterly model (60 points)

Now the standard quarterly calibration:

$$\alpha = 0.33, \quad \beta = 0.99, \quad \delta = 0.025, \quad \ln z' = \rho \ln z + \sigma \varepsilon', \quad \rho = 0.95, \quad \sigma = 0.007,$$

with $c + k' = zk^\alpha + (1 - \delta)k$ and $u = \ln c$. There is **no closed-form policy** for this model — only a closed-form steady state, $k^* = (\alpha\beta/(1 - \beta(1 - \delta)))^{1/(1-\alpha)}$.

- (a) **(15 pts) Where Π comes from.** Implement *both* Tauchen (1986) and Rouwenhorst (1995). For $n \in \{5, 9, 15\}$, report each chain's implied persistence and unconditional standard deviation against the targets ρ and $\sigma/\sqrt{1-\rho^2}$. Then repeat the table at $\rho = 0.99$. Which method would you use, and why?
- (b) **(15 pts) Solve it.** Using the Rouwenhorst chain with $n = 7$, solve the model with your best solver from Question 2. Report the grid, tolerance, sweeps and wall time, and verify that your policy crosses the 45-degree line near the analytical k^* in the middle productivity state. How close is the crossing? If it is further off than one grid spacing, diagnose why — the answer is not that your solver is broken.
- (c) **(15 pts) Simulate and match moments.** Simulate 2,000 quarters (discard the first 500), HP-filter $\ln Y$, $\ln C$ and $\ln I$ with $\lambda = 1600$, and report $\sigma(Y)$, $\sigma(C)/\sigma(Y)$, $\sigma(I)/\sigma(Y)$, $\text{corr}(C, Y)$, $\text{corr}(I, Y)$ and the first-order autocorrelation of Y . Compare with the US post-war targets $\sigma(Y) \approx 1.5\text{--}2.0\%$, $\sigma(C)/\sigma(Y) \approx 0.5\text{--}0.8$, $\sigma(I)/\sigma(Y) \approx 3\text{--}4$, both correlations $\approx 0.8\text{--}0.9$. *An HP filter is provided in Lab 3, Part 7; you may use it or your own.* Then re-run with the *Tauchen* chain and report how much the moments move.
- (d) **(15 pts) Accuracy, and the leaderboard.** Compute unit-free Euler residuals $\mathcal{E} = |1 - \tilde{c}/c|$ at 1,000 capital levels drawn uniformly, *off* the solution grid, across all productivity states. Report $\log_{10}$ of the mean and the maximum, and plot the residual against k . Where are the errors largest, and why?

Submit your maximum $\log_{10} \mathcal{E}$, together with your grid size, tolerance and wall time, for the anonymous class leaderboard.

Bonus (up to 10 points). Endogenous labour

Add a labour choice with $u(c, n) = \ln c - \psi \frac{n^{1+1/\nu}}{1+1/\nu}$ and $y = zk^\alpha n^{1-\alpha}$. Calibrate ψ so that average hours in the simulation are $\bar{n} = 0.3$. Report how $\sigma(Y)$ and $\sigma(I)/\sigma(Y)$ change relative to Question 3, and say which of the target moments the model now gets closer to.

Reporting standard. With every result: the method, the grid (points, spacing, bounds), the chain and its *implied* ρ and σ_z , the tolerance, the dtype, maximum and mean Euler residuals on points not used to solve, and the wall time with the hardware. A number without these is not yet a result.